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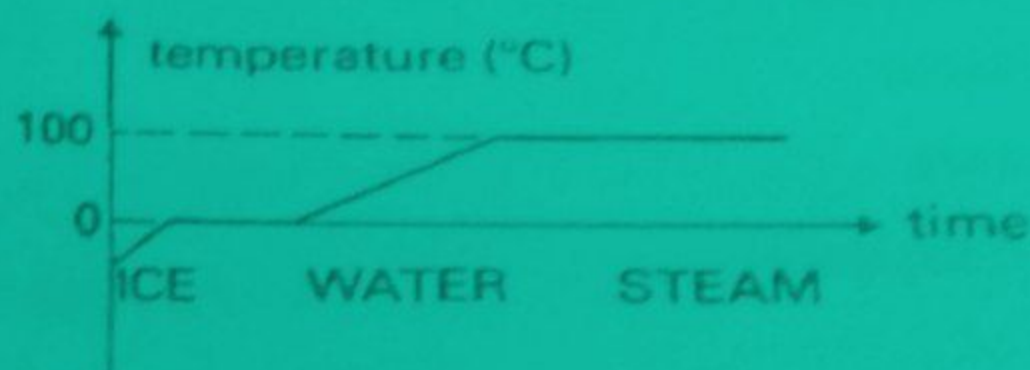
10. A mass is hanging on a string with a spring constant k and simple harmonic motion with a period T . If the mass is doubled to $2m$, the period of oscillation is
- $2T$
 - $T/2$
 - $\sqrt{2}T$
 - $T/\sqrt{2}$
11. A mass attached to a spring vibrates with a period of 4.0 s . When the mass passes through its equilibrium position, it is moving with a velocity of 0.7 m/s . What is the amplitude of the motion?
- 0.016 m
 - 0.073 m
 - 0.1 m
 - 0.4 m
 - 1.25 m
12. Anybody moving with simple harmonic motion is being acted by a force that is
- Constant.
 - Proportional to the inverse square of the displacement.
 - Inversely proportional to the displacement.
 - Proportional to the square of the displacement.
13. The velocity of a particle moving with simple harmonic motion is _____ at the mean position.
- Maximum
 - Minimum
 - Zero
 - None of the above
14. The amplitude of a system moving in simple harmonic motion is doubled. The change in the total energy is _____
- $E \rightarrow 2E$
 - $E \rightarrow 4E$
 - $E \rightarrow 1/2E$
 - $E \rightarrow E$

Part III: Fill in the blank space

- _____ is the flow velocity of the fluid particles at any point is a constant with time.
- A 30 kg hammer strikes a steel spike 2 cm in diameter while moving with speed 25 m/s . The hammer rebounds with speed 10 m/s after 0.1 s . Then the force exerted on the spike during the impact is _____.
- _____ is the amount of energy required to raise the temperature of 1 kg of water by 1°C .



5. Fluid is a substance that
 - A. Cannot be subjected to shear forces
 - B. Always expands until it fills any container
 - C. Has the same shear stress at a point regardless of its motion
 - D. Cannot remain at rest under action of any shear force
6. Bulk modulus is the ratio of _____.
 - A. Increase of pressure and volumetric strain
 - B. Volumetric stress and increase of pressure
 - C. Volumetric strain and increase of pressure
 - D. Increase of pressure and volumetric stress
7. Which one of the following is NOT TRUE about thermo dynamical process?
 - A. A cyclic process is the one that originates and terminates at the same state
 - B. In adiabatic free expansion, then internal energy of the gas does not change.
 - C. An isothermal process is one that occurs at constant temperature
 - D. The work done on a gas during an isobaric process is given by $W = nRT \ln\left(\frac{V_i}{V_f}\right)$
8. Why does the temperature stop rising at 0°C and 100°C in a temperature verses time graph shown below?



- A. The energy goes not into making the molecules to move faster,
 - B. Work has to be done to hold the molecules together,
 - C. The energy goes to separate the molecules from each other,
 - D. All of the above
9. What mass of steam initially at 130°C is needed to warm 200 g of water in a container from 20.0°C to 50.0°C ? ($C_s = 2010 \text{ J/Kg}^\circ\text{C}$ and $C_w = 4186 \text{ J/kg}^\circ\text{C}$)

Part I: Write true if the statement is correct and false if the statement is incorrect [1pt each]

1. Oscillation of simple pendulum is an example of linear oscillation.
2. In isolated system, heat does not exchange and work is done on the environment.
3. If the amplitude of SHM is equal to the position then the velocity is constant.
4. If a weight of a body is more than the upward thrust, then the body sinks.

Part II: Choose the best answer from the given alternatives. [1.5pts each]

1. Which one of the following is not true?
 - A. All points at the same level in a fluid have the same pressure
 - B. Fluid pressure increases in the depth of the fluid
 - C. Atmospheric pressure decreases with increase in altitude
 - D. Fluid pressure depend on the shape of the container
2. Which property of the fluid offers resistance to deformation under the action of shear force?
 - A. Density
 - B. Viscosity
 - C. Permeability
 - D. Specific gravity
3. Which one of the following is not true about ideal fluids?
 - A. The fluid is viscous
 - B. The flow is steady
 - C. The fluid is incompressible
 - D. The flow is irrotational
 - E. None
4. The radius of a water pipe decreases from 10 to 5 cm. If the average velocity in the wider portion is 4 m/s, find the average velocity in the narrower region.
 - A. 8m/s
 - B. 16m/s
 - C. 4m/s
 - D. 12m/s

Part III: Fill in the blank space

4. The change in length of a rod of length 2m with $\alpha = 10^{-6}/^{\circ}\text{C}$, when it is heated through 5K, is _____.
5. The angular position of a pendulum is represented by the equation $\theta = 0.0320 \cos \omega t$, where θ is in radians and $\omega = 4.43 \text{ rad/s}$. Then the period and length of the pendulum respectively are _____ and _____.
6. The total energy of a particle executing SHM of period 2π seconds is $10.24 \times 10^{-3} \text{ J}$. The displacement of the particle at $\pi/4 \text{ s}$ is $0.08 \sqrt{2} \text{ m}$. Then the mass of particle is _____.
7. The heat absorbed or released when matter melts, changing phase from solid to liquid form at constant temperature is _____.
8. On a day when the temperature reaches 50°F , then the equivalent temperature in degree Celsius is _____.
9. A bimetal plate consists of two materials of different coefficients of thermal expansion. The coefficient of thermal expansion of the top part of the plate is less than the bottom part.
 - (a) If the temperature of the entire plate increases then the plate bends _____.
 - (b) If the temperature of the entire plate decreases, then the plate bends _____.
10. A block of wood weights 4N in air and 3N when immersed in liquid. Then the buoyant force on the block wood is equal to _____.

Part IV: Work out problems (10 pts)

1. In an engine, a piston oscillates with simple harmonic motion so that its position varies according to the expression $x(t) = (5.00 \text{ cm}) \cos(2t - \pi/6)$ where x is in centimeters and t is in seconds. At $t = 0$, find
 - a) the position of the piston,
 - b) its velocity
 - c) its acceleration
 - d) the period and
 - e) amplitude of the motion.
2. Calculate the quantity of heat required to raise the temperature of 1 g of ice from -10°C to 110°C .
 $L_v = 2256 \text{ KJ/Kg}$, $L_f = 334 \text{ KJ/Kg}$, $C_s = 2010 \text{ J/Kg}^{\circ}\text{C}$, $C_w = 4186 \text{ J/Kg}^{\circ}\text{C}$ and $C_i = 2090 \text{ J/Kg}^{\circ}\text{C}$

